



THYROID CANCER

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2025 ATA Thyroid Cancer Guidelines: How Has ATA Risk Stratification for Differentiated Thyroid Cancer Evolved?



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KEY POINTS

- The ATA risk stratification, developed to predict disease recurrence rather than mortality, has undergone revision in the 2025 management guidelines for adult patients with differentiated thyroid cancer, incorporating the ongoing accumulation of data to more precisely categorize patients based on risk.
- Key changes include four categories of risk for recurrence (low, low-intermediate, intermediate-high, and high), as well as separate histology types (PTC, FTC and invasive encapsulated follicular variant of PTC, and oncocytic thyroid cancer).
- The new detailed risk of recurrence categories can serve to better inform patient counseling and treatment decisions.

SUMMARY

Background

The AJCC/UICC staging system was designed to predict the risk of death from differentiated thyroid cancer (DTC), but the overall excellent prognosis of DTC led to the development of additional risk-stratification systems focused on recurrence rather than mortality. The 2009 American Thyroid Association (ATA) DTC guidelines introduced a three-tiered

model (low, intermediate, and high risk) to estimate the likelihood of recurrence.¹ In 2015, the ATA guidelines, after incorporating new evidence, substantially refined and expanded this framework.²

Risk stratification plays a pivotal role in guiding initial treatment decisions, which must be tailored to individual patient characteristics. Over the past



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decade, however, emerging data have transformed the understanding of DTC biology and outcomes. The fourth and fifth editions of the WHO Classification of Endocrine and Thyroid Tumors have redefined histopathological taxonomy, including the distinction of oncocytic thyroid carcinoma (OTC) from follicular thyroid carcinoma (FTC) and the categorization of both OTC and FTC into three subtypes, each associated with distinct behavior.^{3,4} All these developments prompted the ATA to update its risk stratification system.⁵

SUMMARY OF NEW RECOMMENDATIONS

The 2025 ATA risk classification introduces several important updates compared with the 2015 system.

1. The traditional three-tier model has been placed by a four-tier classification, comprising:

- Low risk: risk of recurrence: <10%
- Low-intermediate risk: 10–15%
- Intermediate-high risk: 16–30%
- High risk: >30%

This revision aims to reduce the broad heterogeneity of the previous “intermediate” category by subdividing it in two categories.

2. The model now explicitly distinguishes recurrence behavior by tumor histotype:

- Papillary thyroid cancer (PTC) and its subtypes (as defined by the 5th edition of the WHO classification) most commonly present with nodal metastases, but recurrence is rare when disease is intrathyroidal and completely resected.
- Follicular thyroid cancer (FTC) and invasive encapsulated follicular variant of PTC (IEFVPTC) typically spread hematogenously,

with the extent of vascular invasion serving as one of the main risk determinants.

- Oncocytic thyroid cancer (OTC) has a distinct genetic landscape and a higher rate of distant recurrence and presents limited radioactive iodine (RAI) avidity; extensive vascular invasion markedly increases recurrence risk.

3. New features, such as margin involvement, multifocality, tumor size, and extranodal extension have been incorporated into the 2025 classification, refining how recurrence risk is defined:

- Microscopic anterior margin involvement (R1, i.e. the presence of microscopically positive surgical margins) is now included among features of low-risk tumors.
- Microscopic posterior margin involvement (R1) is now included among features of low-intermediate risk tumors.
- Unilateral multifocality is considered a low-intermediate-risk feature in PTC.
- Bilateral multifocality >1 cm is considered an intermediate-high-risk feature in PTC.
- Tumor size >4 cm defines low-intermediate risk tumors, but if associated with microscopic extrathyroidal extension (ETE), it is upgraded to high risk.
- Extranodal extension is newly incorporated as a high-risk feature.

4. The 2025 model emphasizes that the combination of features can have an additive effect on the overall recurrence risk. For instance, the coexistence of two low-intermediate-risk factors reclassifies the tumor as being intermediate-high risk of recurrence. This principle is consistently applied across PTC, FTC, and OTC. Similarly, the combination of specific findings, such as T3a tumors with microscopic ETE, confers a high



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recurrence risk. This underscores that, while the risk of structural disease recurrence exists along a continuum, accumulation of adverse features significantly magnifies overall risk.

5. FTC and OTC, now evaluated separately, are extensively characterized in the 2025 guidelines. The 2015 guidelines reported that FTC with capsular invasion and no or minimal (<4 foci) vascular invasion were low-risk tumors, while FTC with extensive vascular invasion (≥4 foci of vascular invasion) were high-risk tumors. Now, the 2025 guidelines define both FTC and OTC as:

- Low risk: tumor ≤4 cm, minimally invasive (capsular invasion only), no or minimal lymph node (LN) involvement (≤5 LNs, all ≤2 mm),

and negative or microscopic positive anterior margins.

- Low-intermediate risk: tumor >4 cm, or ≤4 cm with limited vascular invasion (<4 vessels), microscopic ETE, cN1a/pN1a >2 mm or involving >5 LNs, or microscopic positive posterior margins.
- Intermediate-high risk: clinically evident lateral LN metastases (cN1b) <3 cm, or coexistence of two low-intermediate-risk factors.
- High-risk: tumor >4 cm with microscopic ETE; T3b or T4 lesions; encapsulated angioinvasive tumors with extensive vascular invasion (≥4 vessels); widely invasive, poorly differentiated, or high-grade histology; presence of extranodal extension, cN1 ≥ 3 cm, or distant metastases.

COMMENTARY

In the recently released ATA guidelines on the management of adult patients with DTC, major steps have been taken to increase the granularity of recurrence risk assessment, with two main features being the separate classification of intermediate risk of recurrence tumors into either low-intermediate or intermediate-high risk and the subclassification of DTC into PTC and its subtypes, FTC/IEFVPTC, and OTC. In doing so, the guidelines provide the most comprehensive framework to date to help clinicians associate a tumor's pathologic features with a percentage range for risk of clinical recurrence, which in turn, facilitates patient-clinician discussions and decision-making with regard to radioactive iodine (RAI) therapy. Shared features associated with high risk of recurrence across all DTC subtypes include tumor status of T3a and higher, extra-thyroidal extension, aggressive histology, clinical nodal involvement >3 cm, presence of extranodal extension, and distant metastasis. Unique to FTC,

IEFVPTC and OTC, widely invasive tumors with involvement of ≥4 vessels are considered high risk of recurrence. Another important change is that the co-occurrence of two or more risk factors in the low-intermediate-risk category “upgrades” DTC in the intermediate-high group, which reflects how recurrence risk is more of a continuum than a binary classification system. Ultimately, risk stratification for DTC is a key tool closely tied to decision-making regarding RAI administration after thyroidectomy. Despite the two subgroups of intermediate risk of recurrence, this does not translate in different RAI recommendations, which is typically considered for any intermediate-risk disease, for remnant ablation and potentially adjuvant therapy, at activity levels between 30 and 100 mCi. On the other hand, typically RAI is not recommended for low-risk DTC, and it is recommended for high risk of recurrence DTC at activity levels between 100 and 150 mCi.



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The 2025 guidelines are definitely comprehensive in characterizing risk of clinical recurrence, yet the amount of information and detail is large and complex to absorb, particularly for clinicians practicing general endocrinology. It will take some time and ongoing education for the endocrine community to become familiarized with the all-encompassing

content on DTC risk stratification and to successfully incorporate new guideline recommendations in their clinical practice.

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REFERENCES

1. Cooper DS, Doherty GM, Haugen BR, et al. Revised American Thyroid Association management guidelines for patients with thyroid nodules and differentiated thyroid cancer. *Thyroid* 19:1167–1214; doi: 10.1089/thy.2009.0110.
2. Haugen BR, Alexander EK, Bible KC, et al. 2015 American Thyroid Association management guidelines for adult patients with thyroid nodules and differentiated thyroid cancer: the American Thyroid Association guidelines task force on thyroid nodules and differentiated thyroid cancer. *Thyroid* 2016;26:1-133; doi: 10.1089/thy.2015.0020.
3. Lloyd R, Osamura R, Kloppel G, Rosai J, eds. 2017 WHO Classification of Tumours of Endocrine Organs, 4th ed. Lyon, France: IARC Publications, 2017.
4. Ghossein R, Sobrinho-Simoes M, Tallini G, Asa S, eds. 2022 WHO Classification of Tumours Endocrine and Neuroendocrine Tumours, 5th ed. Lyon, France: IARC Publications; 2022.
5. Ringel MD, Sosa JA, Baloch Z, et al. 2025 American Thyroid Association management guidelines for adult patients with differentiated thyroid cancer. *Thyroid* 2025;35(8):841-985; doi: 10.1177/10507256251363120.

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